

SUMMARY

Machine Learning Engineer specializing in production **multi-agent systems** and enterprise **GenAI** applications. Reduced customer support resolution times by 45% through intelligent automation workflows and architected real-time analytics platforms processing 10K+ data points with sub-2-second latency. Expert in **LangChain/LangGraph** orchestration, **RAG** pipelines, and deploying **LLM**-powered solutions on Cloud infrastructure (**AWS, GCP**).

SKILLS

- **Languages:** Python, SQL, JavaScript, C++.
- **Frameworks/Tools:** LangChain, LangGraph, OpenAI API, CrewAI, Hugging Face Transformers, RAG, AutoGen, LlamaIndex, Haystack
- **Machine Learning:** TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, OpenCV, RLHF.
- **LLMs:** GPT, Claude, LLaMA, Mistral, Gemini, OpenAI Whisper, Deepseek.
- **Web and Technologies:** FastAPI, RESTful APIs, WebSockets, React, Node.js, Unix shell.
- **Databases:** PostgreSQL, SQL, MySQL, MongoDB, FAISS, Elastic Search, Pinecone, ChromaDB.
- **Visualization:** Streamlit, Power BI, Tableau
- **Cloud/DevOps:** AWS (Lambda, S3, EC2, Timestream, DynamoDB, Amplify), GCP, Docker, Kubernetes, GitHub Actions.
- **Others:** Git, CI/CD, MLOps, Pyspark, Kafka, Snowflake, RAG pipelines, Agentic orchestration, Prompt Engineering.

EXPERIENCE

Balance AI, USA | Software Developer & GenAI Engineer

Sept 2025 – Current

- Architected an intelligent analytics chatbot using LangGraph **multi-agent orchestration**, processing 10K+ sensor data points per query with **<2s** response latency across temperature, presence, and environmental monitoring use cases.
- Integrated **Tuya Cloud APIs** with AWS serverless infrastructure (**Lambda, Timestream, DynamoDB**) to enable real-time streaming analytics, reducing data retrieval overhead by 60% through optimized query patterns and caching strategies.
- Developed **autonomous agent** logic for statistical computations (correlation analysis, trend detection, anomaly identification) with dynamic visualization selection, delivering actionable insights that identified 15-35% potential energy optimization opportunities.
- Deployed production-ready **Next.js** frontend into **AWS Amplify** with real-time streaming responses, state persistence, and conversational memory—achieving 95% user query success rate in beta testing.

AT&T, USA | Machine Learning & GenAI Engineer

Sep 2024 – July 2025

- Engineered AI pipelines for intelligent call routing using **LangGraph and LangChain** multi-agent frameworks with Model Context Protocol (**MCP**) integration, enhancing customer support automation and reducing resolution times by 45%.
- Created **RAG**-based GenAI applications that integrated **vector stores** (FAISS, Pinecone) with **OpenAI** embeddings for **semantic search** and document Q&A across 10K+ enterprise documents.
- Built fairness-aware training pipelines for **NLP systems**, reducing flagged incidents in compliance testing by 30%.
- Designed **multi-agent** orchestration frameworks and automation pipelines for customer service and fraud detection workflows.
- Fine-tuned **GPT-4** and **LLaMA2** models with **QLoRA** and **LORA**, improving response relevance by 27% in domain-specific knowledge systems.
- Built interactive web tools using **FastAPI** and **LangChain** for HR, Finance, and Legal departments to query internal knowledge bases with natural language.

Symantrix, India | ML Engineer

Aug 2020 – Aug 2022

- Implemented **NLP** pipelines for document classification, named entity recognition, and semantic similarity using BERT and spaCy.
- Transitioned rule-based logic into classical ML pipelines for automated email triage and document tagging.
- Developed interactive chatbot frontends with **React-JavaScript** for user-facing applications powered by machine learning models.
- Collaborated with data engineers to automate data pipelines and model training workflows on AWS (S3, EC2, SageMaker).
- Built supervised and unsupervised ML models (**Random Forest, K-Means, XGBoost**) to detect anomalies and forecast sales.

Learnasky, India | Co-Founder, Full Stack Developer

May 2019 – Aug 2020

- Co-founded a startup and developed a web application using the MERN stack for assignment posting and submission, improving academic workflow efficiency by 30%.
- Architected **MongoDB** schemas and built **RESTful APIs** with **Node.js** and React to optimize data storage and reduce latency by 15%.

EDUCATION

Master of Science in Computer Science.

Aug 2022 – May 2024

Colorado State University, USA

Bachelor of Technology in Computer Science and Engineering

Aug 2016 – May 2020

Jawaharlal Nehru Technological University, Hyderabad (JNTUH), India

TECHNICAL PROJECTS

Multi-Agent AI for Safe Dining: Menu Intelligence and Dietary Filtering

- Built a **multi-agent** dietary assistant using **LangGraph and GPT-4o** that analyzes restaurant menus with 90%+ accuracy for allergen detection across vegan, gluten-free, and dietary restrictions, leveraging planner, worker, and evaluator agent architecture with real-time web scraping via Playwright.

Autonomous AI Interview Platform with Code Evaluation and Voice Interaction

- Designed an AI-driven interview assistant with autonomous agents for dynamic DSA question delivery, voice interaction, and real-time code evaluation across multiple programming languages. Integrated **OpenAI Whisper** into a **Flask-based** backend to enable accurate **speech-to-text transcription**, powering voice-controlled agent interactions.

Research Project: Misinformation Detection System

Aug 2023 – Dec 2023

- Designed and implemented an **NLP**-based fake news classifier using **GloVe embeddings and LSTM**, achieving up to 90.98% accuracy after conducting comprehensive evaluations across various **vectorizers and classifiers**, including **BERT, CNNs**, and traditional machine learning/Deep Learning models.